European Space Policy Institute

EXECUTIVE SUMMARY

China is one of the world's key players in the development of new advanced technologies, including artificial intelligence (AI), robotics, and other digital technologies. To continue its leadership and to satisfy the demands for large-scale compute, China has now become one of the biggest proponents and early-adopters of orbital data centres (ODCs).

In its approach, it has homed in on four elements: (1) top-down policy drafting and guidance, (2) private and state-owned industry adaptation, (3) municipal and provincial government incentivisation, and (4) supporting non-state actors and groups. Central to this effort have been the state-owned China Aerospace Science and Technology Corporation (CASC), who first formalised ODCs in a new strategy; the Beijing Municipal Government, and the Zhongguancun Science Park in Beijing; as well as a multitude of startups.

Timeline of China's ODC activities, 2025-2026



Source: Based on the authors' own compilation.

As China pursues orbital compute capabilities, the technological gap widens. Europe risks having to compute its (space) data through third-party actors as well as lacking access to orbital information. To ensure Europe's technology sovereignty is future-proof, this policy brief offers the following recommendations:

1. The EU should establish coordinated political guidance, investment support and regulatory framework for ODC developments.
2. The Horizon Europe Moonshot Projects under the 2028-2034 Multiannual Financial Framework (MFF) should include a flagship European ODC initiative.
3. ESA and EU should signal demand and offer associate contracts for in-space data processing for start-ups and scale-ups.
4. European actors should incentivise start-ups that address current feasibility challenges through a European ODC Challenge.
5. European space actors should convene policy and technology working groups with lead actors and companies worldwide on ODC developments and standards.